D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Discuss the current status of DNA microarray profiling in management of malignant tumour.

February 2022 | DM 0222 | Q1 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction & exam framing (Feb 2022 essay)

DNA microarray gene-expression profiling (GEP) measures simultaneous mRNA abundance of thousands of genes on a solid substrate. In oncology it moved from unsupervised discovery of molecular subtypes to **clinically validated prognostic and predictive signatures**, especially in early breast cancer.

A high-scoring answer should (1) explain the technology and QC, (2) contrast prognostic vs predictive use, (3) detail Oncotype DX, MammaPrint, PAM50/Prosigna (and related assays), (4) note GEP in myeloma/ALL/lymphoma with limitations, (5) compare microarray vs RNA-seq/NGS, and (6) close with Indian/NCG practicality and NCCN-aligned breast pathways.

- Definition one-liner: simultaneous transcriptome snapshot → risk score or subtype.
- Clinical maturity is highest in ER+/HER2– early breast cancer; elsewhere mostly research or specialized centers.
- Always state: signature ≠ substitute for stage, grade, receptor status, or germline testing.

EXAM PEARLS / HALLMARKS

- Open with: ‘Microarrays classified biology; selected multigene assays now guide adjuvant chemo omission.’
- Examiners expect TAILORx and MINDACT numbers — write at least one HR or absolute DMFS/%.

2. Technology: platforms, workflow & quality control

Classic platforms: spotted cDNA arrays and high-density oligonucleotide arrays (e.g., Affymetrix GeneChip). Modern clinical assays are often RT-PCR or RNA-based kits derived from microarray discovery cohorts, run on FFPE with standardized RNA extraction.

Workflow: tissue adequacy → RNA isolation → reverse transcription / labeling → hybridization → scanning → normalization → signature algorithm → clinical report. Pre-analytics dominate error: ischemic time, fixation, tumor cellularity, and necrosis.

- Minimum tumor cellularity thresholds are assay-specific (often ≥10–30% invasive tumor).
- Housekeeping-gene normalization and batch-effect correction are mandatory for research arrays.

• Clinical assays report a locked algorithm (no ad-hoc re-clustering at the bedside).

Microarray / multigene assay clinical workflow

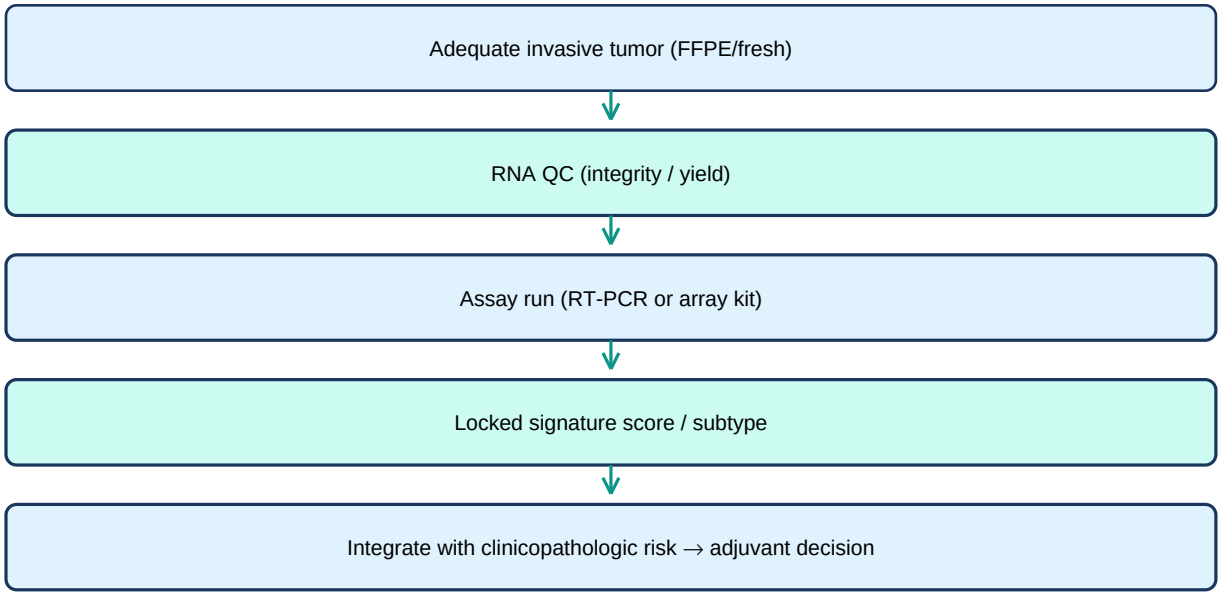


Figure: stepwise exam algorithm

mRNA from tumor	Reflects active transcription programs
Probe hybridization	Quantifies relative gene expression
Normalization	Removes technical noise
Signature score	Maps to prognosis / chemo benefit
Clinical action	Escalate or omit adjuvant therapy

Figure: mechanism / pathway map

3. Biology captured by expression profiling

GEP recapitulates pathways of proliferation (e.g., Ki-67–related genes), ER signaling, HER2 amplicon activity, immune infiltration, and basal programs. Unsupervised clustering historically defined intrinsic breast subtypes: Luminal A/B, HER2-enriched, Basal-like, Normal-like.

Important exam nuance: IHC surrogates approximate but do not perfectly match intrinsic subtypes; PAM50 is the gene-expression gold-standard for intrinsic subtype assignment in many studies.

• Proliferation modules drive much of the prognostic power in ER+ disease.

• Immune gene signatures correlate with CPI benefit in some tumors — still largely investigational for microarray clinical kits.

• Expression ≠ mutation: microarray will not replace DNA NGS for EGFR/BRAF/BRCA targets.

Proliferation genes	Higher → higher recurrence risk
ER module	Endocrine sensitivity correlate
HER2 module	HER2-enriched biology
Basal / immune modules	Aggressive / IO-relevant biology

Figure: mechanism / pathway map

4. Oncotype DX (21-gene RS) — the exam workhorse

Oncotype DX quantifies 16 cancer-related + 5 reference genes by RT-PCR on FFPE, yielding Recurrence Score (RS) 0–100. It is both **prognostic** (distant recurrence on endocrine therapy) and **predictive** of chemotherapy benefit in ER+/HER2– early breast cancer.

Historical cutoffs (NSABP B-14/B-20 era): low <18, intermediate 18–30, high ≥31. Prospective TAILORx redefined clinical bands used today: RS 0–10, 11–25, and ≥26.

- Best validated in node-negative and limited node-positive (1–3) HR+/HER2– disease.
- Does not replace HER2 testing or staging; HER2+ disease is outside classic RS chemo-sparing paradigms.
- Report turnaround and cost matter for Indian practice — discuss affordability and selective use.

Trial	Key result / exam takeaway
NSABP B-20 (retrospective)	High RS derives large chemo benefit; low RS little benefit
TAILORx (NEJM 2018)	N0, RS 11–25: endocrine alone non-inferior to chemo→endocrine for iDFS overall
TAILORx age subgroup	Women ≤50 y with RS 16–25: chemo may reduce distant recurrence
RxPONDER (NEJM 2021)	N1 (1–3 nodes), RS ≤25: postmenopausal — no chemo benefit; premenopausal — chemo benefit

Table: landmark clinical trials

EXAM PEARLS / HALLMARKS

- Memorize: TAILORx = N0; RxPONDER = N1 — menopausal status flips the chemo decision at intermediate RS.

5. TAILORx & RxPONDER — numbers to quote

TAILORx: HR+/HER2–, N0, RS 11–25 randomized to endocrine vs chemo + endocrine. Invasive disease-free survival was non-inferior for endocrine alone in the overall intermediate group (e.g., 9-year iDFS ~83% both arms in primary report). Absolute chemo benefit emerges mainly in younger women with mid-range RS.

RxPONDER: 1–3 positive nodes, RS ≤25. Postmenopausal: 5-year iDFS similar with or without chemo (~91–92%). Premenopausal: chemo improved iDFS (≈ absolute ~3–5%+ depending on follow-up cut) and distant relapse-free intervals — OFS strategies are discussed as alternative contributors in debates, but trial as reported supports chemo benefit.

- RS ≥26 (N0) or high RS with nodes: chemotherapy generally recommended unless contraindicated.
- RS ≤10 (N0): endocrine therapy alone — excellent prognosis.
- Always integrate anatomic stage, grade, LVI, comorbidity, and patient preference.

Oncotype-guided adjuvant chemo decision

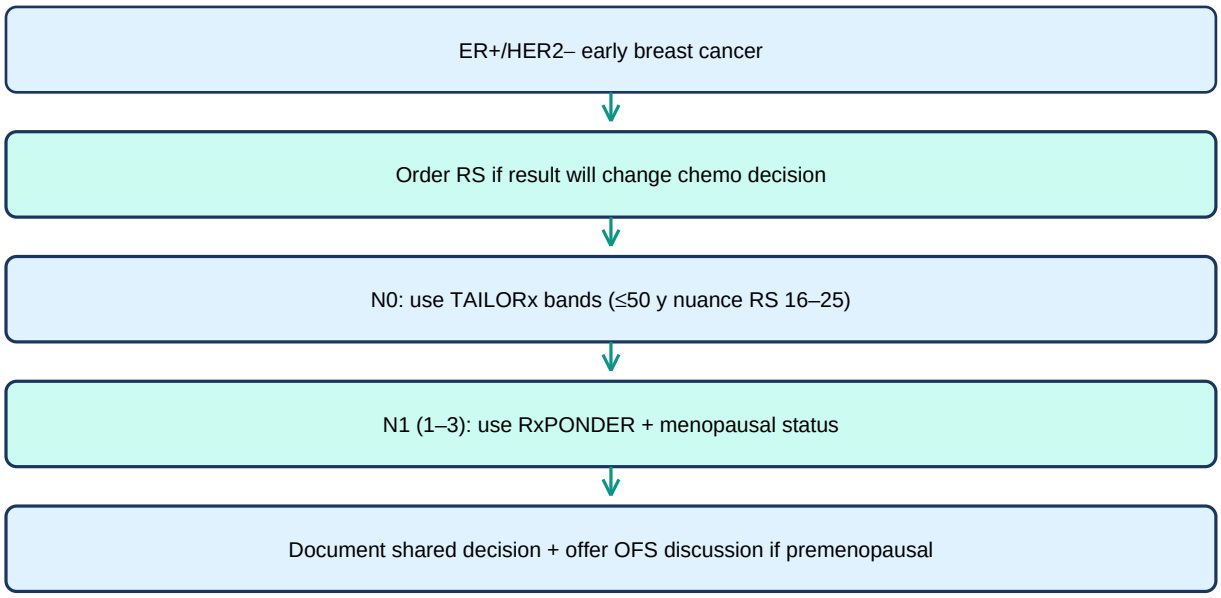


Figure: stepwise exam algorithm

6. MammaPrint (70-gene) & MINDACT

MammaPrint (Amsterdam 70-gene signature) dichotomizes genomic High vs Low risk. Originally fresh tissue / microarray; FFPE versions exist. **MINDACT** (NEJM 2016; long-term updates) used clinical risk (Adjuvant! Online) × genomic risk.

Key practice point: clinically high risk / genomically low risk patients who omitted chemotherapy had 5-year distant metastasis-free survival ≈ 94% — supporting chemo sparing in selected patients. Absolute chemo benefit in discordant groups is small; shared decision-making is essential.

- MammaPrint = binary risk; Oncotype = continuous RS with predictive chemo data in node± settings.
- Do not mix cutoffs across assays — they are not interchangeable.

Trial	Key result / exam takeaway
MINDACT primary	Clinical high / genomic low: chemo omission feasible; 5-y DMFS ~94.7% without chemo (per primary analysis framing)
MINDACT long-term	Confirms limited absolute chemo benefit in genomic-low discordant patients
Microarray discovery (van't Veer / van de Vijver)	70-gene signature prognostic in early breast cancer

Table: landmark clinical trials

7. PAM50 / Prosigna, EndoPredict, BCI & comparisons

PAM50 intrinsic subtype + Risk of Recurrence (ROR) score (Prosigna) provides prognostic information in HR+ early breast cancer and is used in some guidelines/centers for late-distant recurrence risk. **EndoPredict** (12-gene; EPclin combines clinical factors) and **Breast Cancer Index (BCI)** help prognosis and, for BCI, decisions on extended endocrine therapy.

Comparative pearl: Oncotype has the strongest prospective predictive evidence for adjuvant chemo benefit (TAILORx/RxPONDER). Others are primarily prognostic or endocrine-duration tools.

- Prosigna: intrinsic subtype (LumA/LumB/HER2-E/Basal) + ROR.
- BCI: prognostic + predictive for extended AI benefit after 5 years endocrine therapy in selected N0/N1.

• IHC4 / clinical models are cheaper alternatives when genomic assays unavailable (NCG resource-stratified thinking).

Trial	Key result / exam takeaway
TransATAC / ABCSG cohorts	PAM50 ROR, EndoPredict, BCI prognostic validation sets
BCI extended endocrine analyses	Identifies who benefits from years 5–10 endocrine therapy

Table: landmark clinical trials

8. Beyond breast: myeloma GEP, ALL, lymphoma COO

Multiple myeloma: GEP70/GEP80 (UAMS) and related signatures stratify high-risk biology (e.g., proliferation, chromosome 1q–linked programs). They refine prognosis beyond ISS/R-ISS but are not universally mandated; NGS/FISH for del(17p), t(4;14), t(14;16), 1q gain remain standard.

ALL: expression profiling helped define Ph-like ALL and other subtypes historically; contemporary diagnosis leans on NGS fusions, CRLF2, and flow/genomics rather than clinical microarray kits.

DLBCL cell-of-origin: microarray GCB vs ABC (Wright/Alizadeh) → IHC algorithms (Hans) as surrogates; NanoString Lymph2Cx used in trials. COO alone no longer drives most frontline choices after failed subtype-selected trials with R-CHOP ± novel agents in unselected practice — nuance for exams.

• Myeloma: quote GEP as research/prognostic adjunct — treatment still driven by IMWG risk + frailty + induction trials (e.g., anti-CD38 quadruplets).

• ALL: Ph-like identification matters for TKI/trial options — method is usually NGS/MLPA, not bedside microarray.

• Lymphoma: COO is biologic literacy; PET-adapted and molecular subtypes (e.g., MCD, BN2) are evolving.

EXAM PEARLS / HALLMARKS

• Say explicitly: ‘GEP in myeloma/ALL informs biology; routine Indian practice still FISH/cytogenetics/NGS-first.’

9. Other solid tumors — status check (honest limitations)

Colon Cancer: ColoPrint and similar GEP assays remain less practice-changing than MSI/RAS/BRAF and ctDNA MRD. Prostate: Decipher genomic classifier (expression-based) has growing role in post-prostatectomy risk and ADT intensification debates. Lung and others: expression signatures for IO prediction underperform relative to PD-L1, TMB, and driver NGS.

Bottom line for the essay: microarray profiling’s **clearest management impact** remains adjuvant breast cancer chemo and endocrine-duration decisions.

• Do not oversell pan-cancer microarray panels as standards of care.

• DNA NGS actionable mutations > expression arrays for targeted drug selection in advanced disease.

10. Microarray vs RNA-seq vs DNA NGS — choose the right tool

Microarrays quantify known probe sets; RNA-seq offers wider dynamic range, novel transcripts, and fusion detection. DNA NGS detects mutations, copy number, MSI, TMB. Clinical multigene ‘recurrence scores’ may use RT-PCR on a fixed gene list — conceptually descendants of microarray discovery.

• Predictive targeted therapy → DNA/RNA fusion NGS + IHC.

• Adjuvant chemo sparring in ER+ breast → validated expression signatures.

• Minimal residual disease → ctDNA, not microarray.

DNA microarray / RT-PCR signature	Fixed genes → risk score (Oncotype/MammaPrint era)
RNA-seq	Discovery, fusions, broader transcriptome
DNA NGS panel	Actionable mutations / CNAs / MSI
IHC / FISH	Protein/amplification bedside standards

Figure: mechanism / pathway map

11. Limitations, pitfalls & ethics

Limitations: assay discordance, cost, turnaround, limited predictive breadth outside breast, racial/ethnic under-representation in some training sets, and risk of over-reliance ignoring clinical high-risk features. Heterogeneity and biopsy site bias can misclassify.

Ethics: financial toxicity and equity — offer assay when it materially changes therapy; otherwise use clinicopathologic models.

- Never omit chemo solely on genomic low risk if inflammatory breast cancer or other extreme clinical risk without supportive data.
- Repeat testing rarely indicated; prefer correct initial indication.
- Counsel that intermediate results still need judgment (especially premenopausal).

12. NCCN / guideline position (breast-focused) & 2026 framing

NCCN Breast Cancer pathways endorse consideration of validated multigene assays to assess prognosis and potential chemo benefit in HR+/HER2− early disease with specified anatomic stages. Paper IV answers should cite **NCCN Breast Cancer v9.2026** currency alongside classic trial evidence.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Breast Cancer v9.2026; NCCN Guidelines Navigator (breast)
- Use 21-gene RS (or other validated assays per NCCN footnotes) to guide adjuvant systemic therapy in appropriate HR+/HER2− patients.
- Integrate menopausal status with RS (TAILORx/RxPONDER logic) when discussing chemo.
- NCCN 2026 metastatic HR+ additions (e.g., imlunestrant for ESR1) are separate from adjuvant GEP — mention only to show guideline literacy.
- Genomic assays complement, never replace, AJCC staging and receptor testing.

13. Indian / NCG practical notes

In many Indian centers, selective Oncotype/MammaPrint use is constrained by cost. NCG resource-stratified approaches emphasize accurate ER/PR/HER2/Ki-67, anatomic stage, and MDT discussion. When assay unavailable: offer chemo for higher anatomic/biological risk; consider omission in unequivocally low clinical risk postmenopausal N0 disease after counseling.

- Prioritize funding assays for borderline cases where decision truly hinges on genomic risk.
- Ensure receptor re-testing quality (ASCO/CAP) before expensive GEP.
- Document financial counseling in the file — exam-friendly phrase.

14. Future directions

RNA-seq signatures, spatial transcriptomics, and multimodal AI (pathology + expression + radiology) are research frontiers. ctDNA MRD may overshadow static GEP for escalation/de-escalation in coming years. Tumor-agnostic DNA biomarkers already dominate metastatic matching.

- Watch prospective trials combining MRD + GEP.
- Immune GEP for CPI selection — promising, not yet microarray-kit standard across cancers.

15. Summary algorithm & exam pearls

DNA microarray profiling established the molecular taxonomy of cancer; clinically, locked multigene expression assays — especially Oncotype DX and MammaPrint — meaningfully guide adjuvant therapy in HR+/HER2– breast cancer. GEP elsewhere is adjunctive. Pair every score with clinicopathologic context, menopausal status, and patient values.

When microarray-derived assays change management

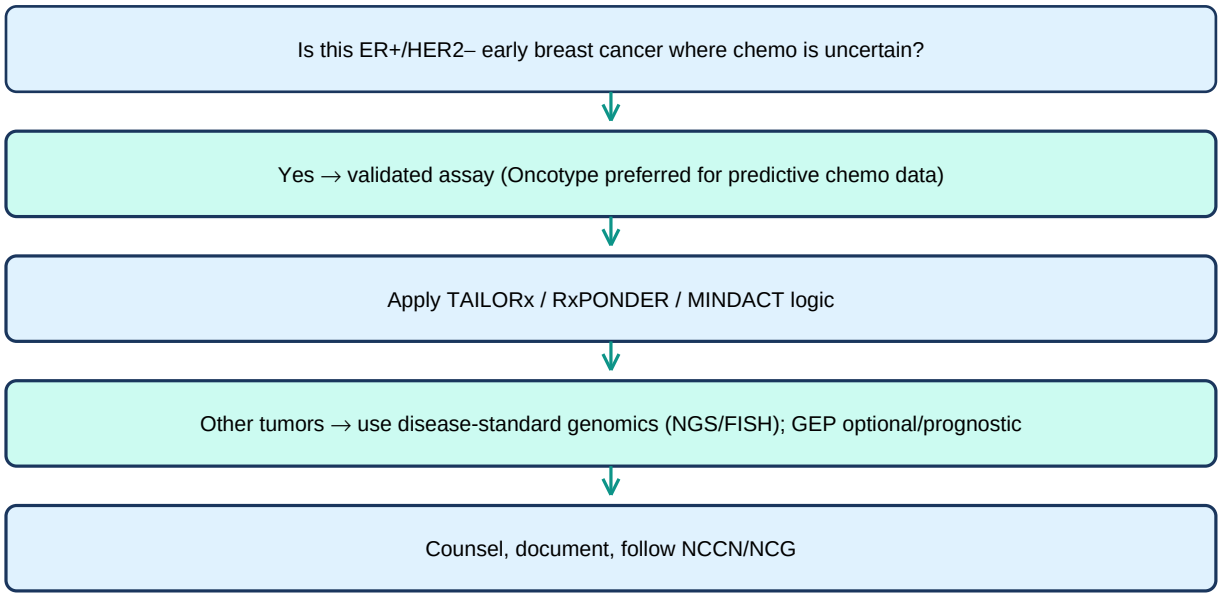


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- TAILORx = N0 intermediate RS; RxPONDER = N1 RS≤25 + menopause split.
- MINDACT: clinical high / genomic low → chemo omission often acceptable.
- Myeloma GEP high-risk ≠ mandatory assay in all Indian clinics — know FISH first.
- Microarray discovers; RT-PCR kits deliver; NGS targets drugs.
- Write one limitation paragraph — examiners reward maturity.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.

- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

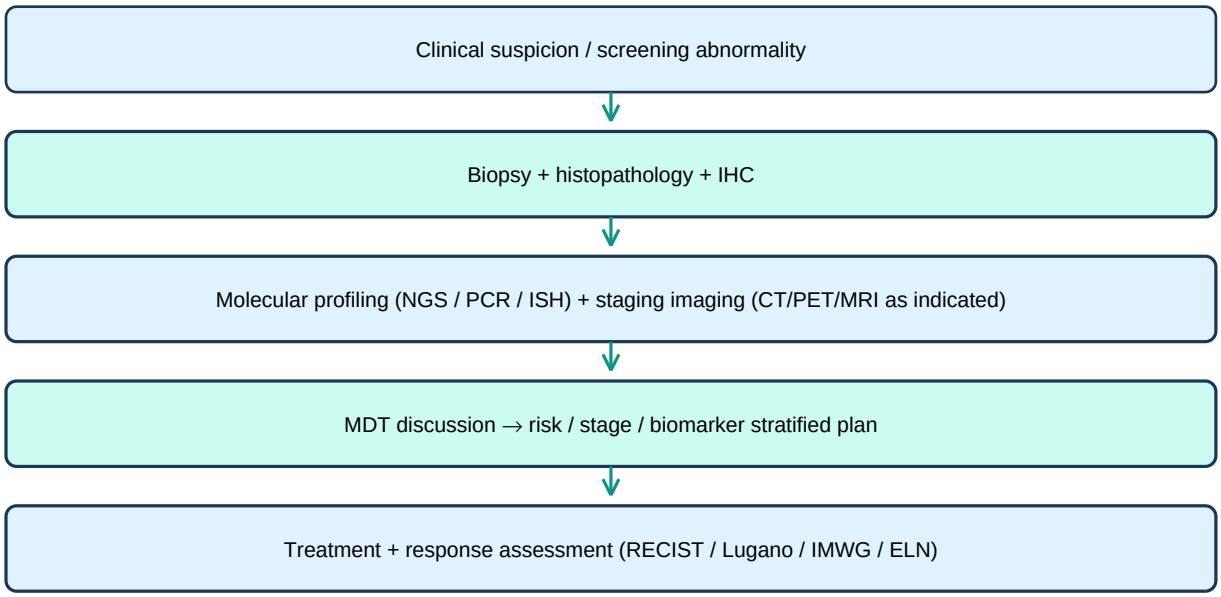


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

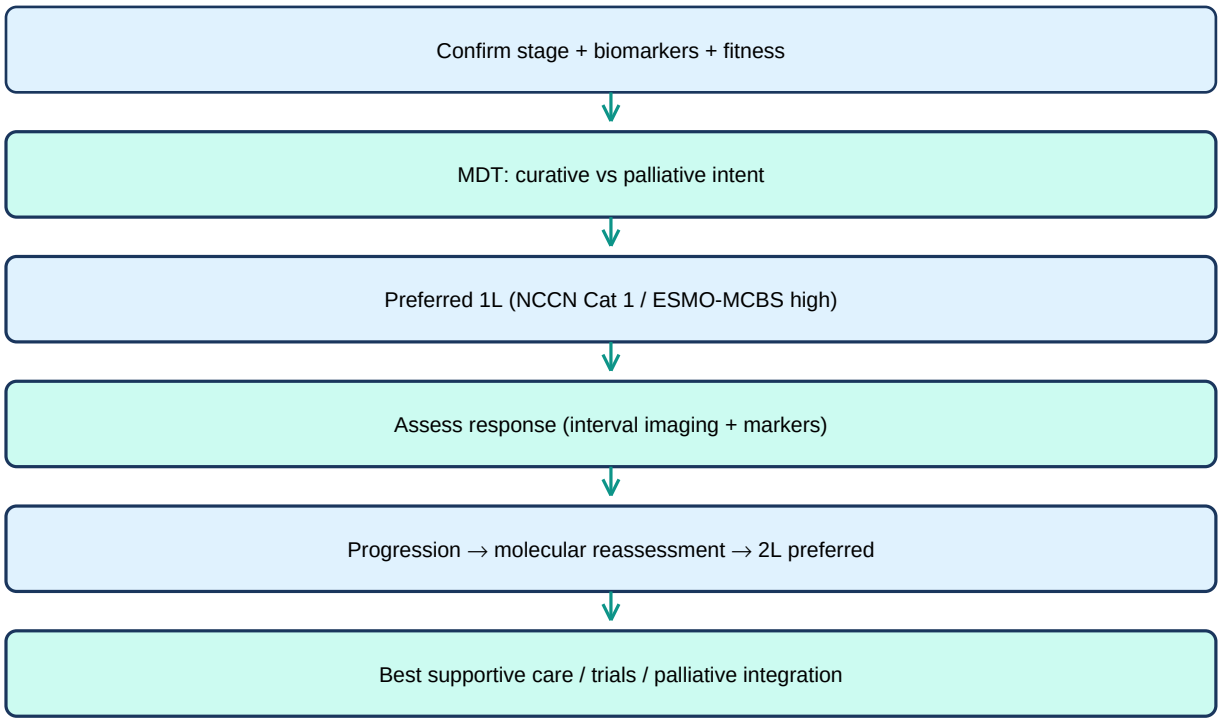


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

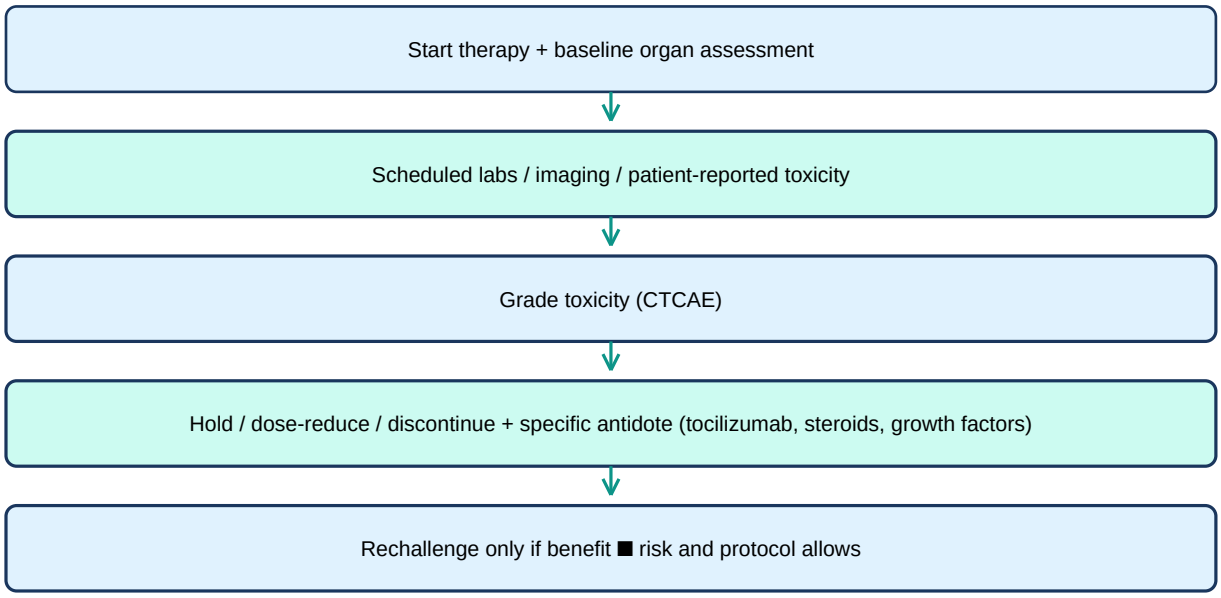


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* NCCN Guidelines® (latest 2026/2027 disease-specific version)
- Cite the latest disease-specific NCCN Guidelines® (2026/2027 versions) and NCCN Drugs & Biologics Compendium for category ratings.
- Mention Category 1 vs 2A/2B and preferred vs other recommended vs useful in certain circumstances.
- Cross-check FDA label, institutional availability (India), and National Cancer Grid pathways where relevant.

EXAM PEARLS / HALLMARKS

- NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).
- Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

- MDT documentation improves outcomes and exam completeness.
- Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.
- Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

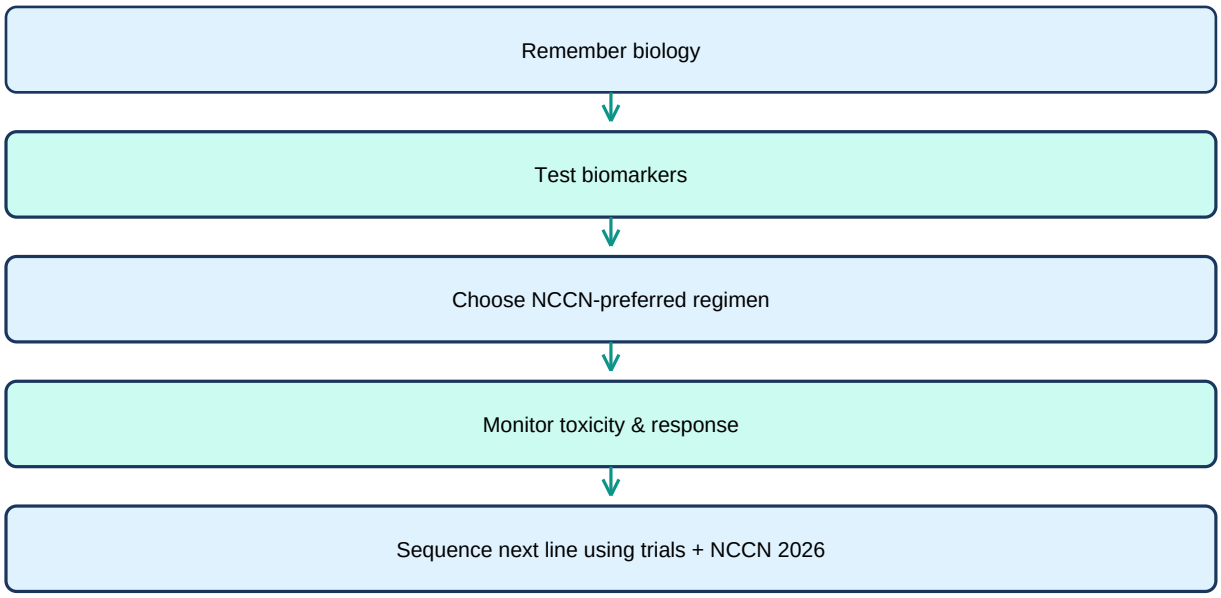


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.

• ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

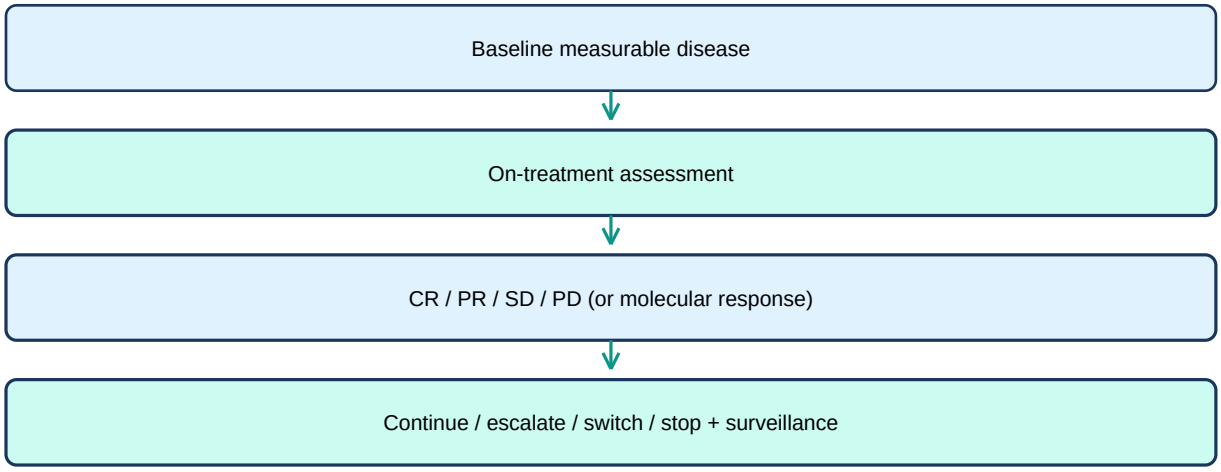


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

• Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.

• Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

• Document MDT decisions; clinical trial referral when standard options exhausted.

• Counsel financial toxicity transparently; prioritize highest-value Category 1 options.

• Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

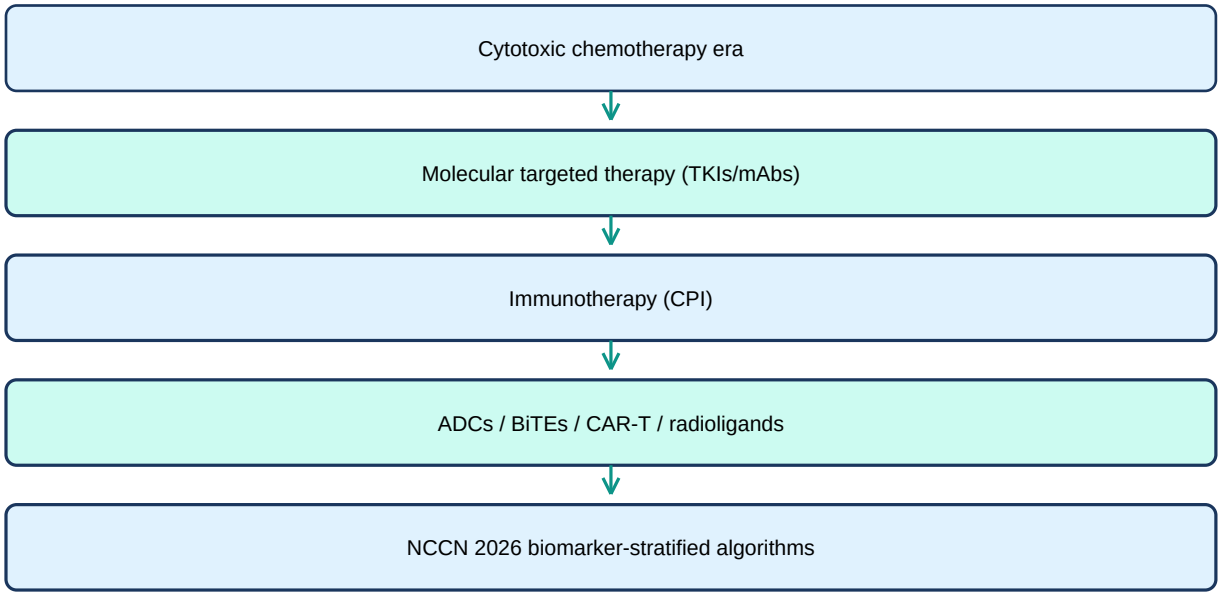


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

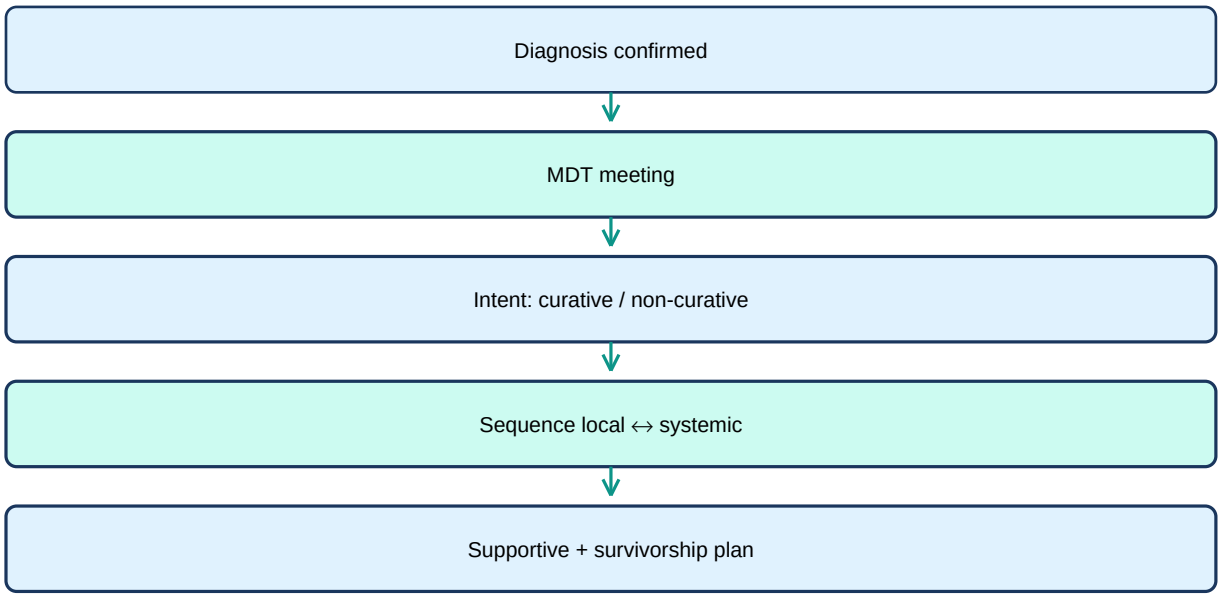


Figure: stepwise exam algorithm

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome

in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

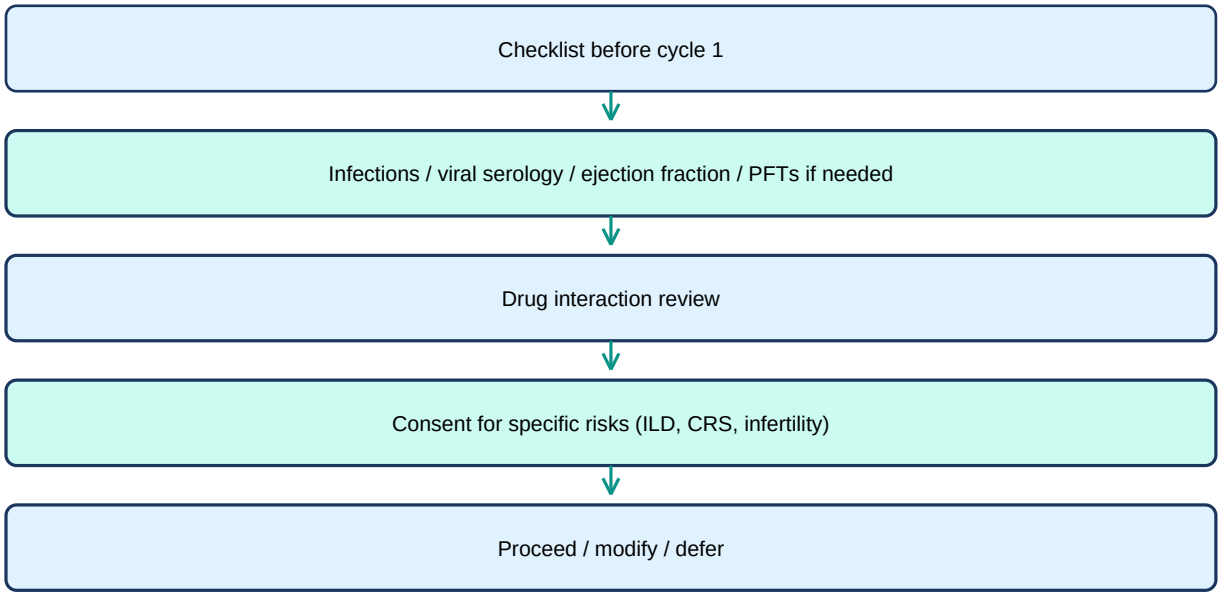


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write 'NCCN 2026' in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* NCCN Guidelines® (latest 2026/2027 disease-specific version)

• Cite the latest disease-specific NCCN Guidelines® (2026/2027 versions) and NCCN Drugs & Biologics Compendium for category ratings.

• Mention Category 1 vs 2A/2B and preferred vs other recommended vs useful in certain circumstances.

• Cross-check FDA label, institutional availability (India), and National Cancer Grid pathways where relevant.

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

• If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.

• Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

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Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

NCCN 2026 / 2027 UPDATES (include in answer)

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Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.